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Lab 3: Component Modelling & Architectural Pattern Selection

Project: Faculty Research Grant & Publication Tracker

Introduction

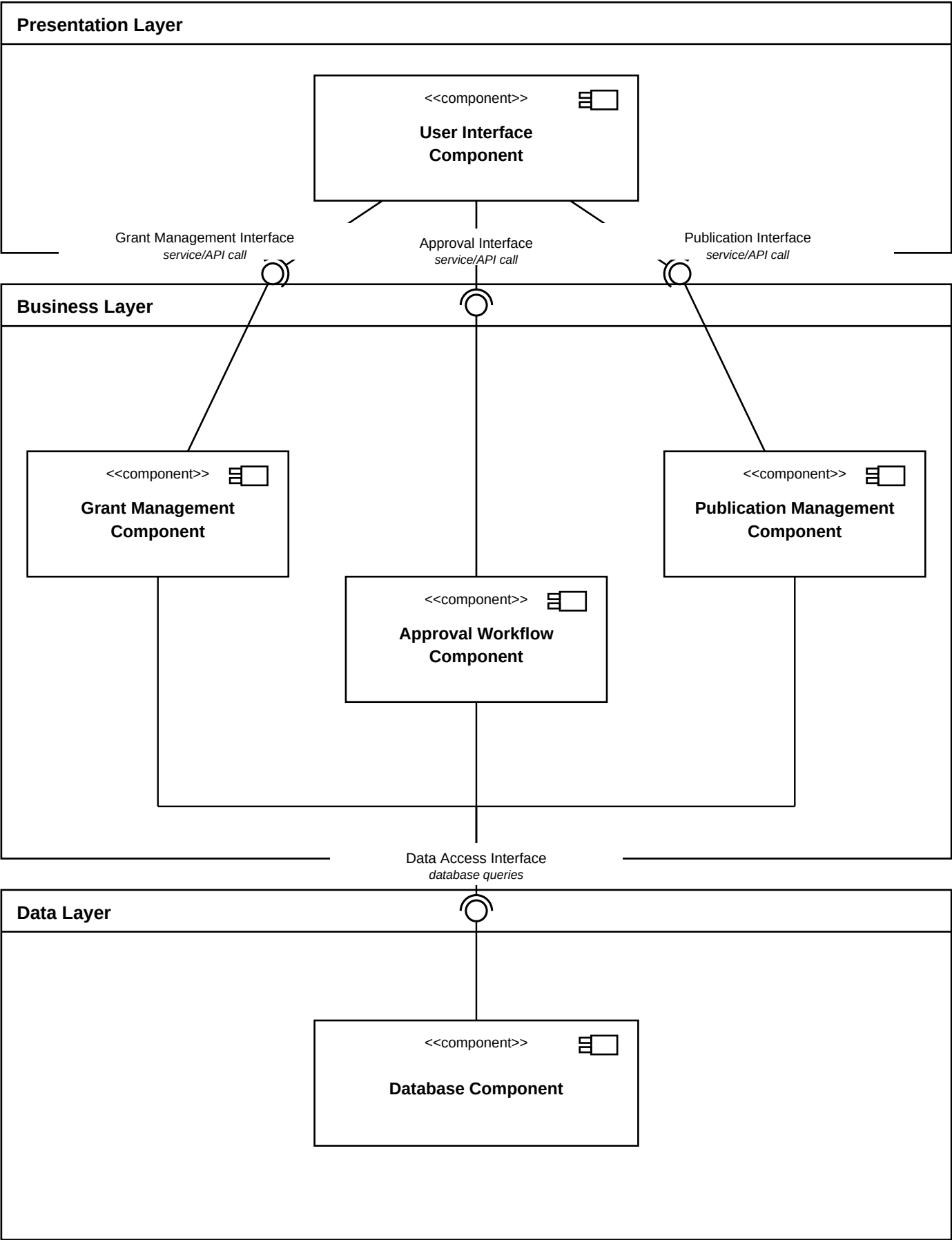
The Faculty Research Grant & Publication Tracker is a centralized academic portal for managing research grants, publication information, approval workflows, and grant fund analytics. The system is used by faculty researchers, co-authors, and the Research Dean to record and review research-related information while maintaining secure and reliable access to stored data.

Selected Architectural Style: Layered Architecture

The system is organized into a Presentation Layer, a Business Layer, and a Data Layer. This structure separates user interaction, application logic, and data storage while keeping the system simple to understand and maintain.

UML Component Diagram

Faculty Research Grant & Publication Tracker



Architectural Justification

Architecture Selection

We chose Layered Architecture for the Faculty Research Grant & Publication Tracker System.

Reason 1 - Separation of Responsibilities

The system contains distinct functions for grant management, publication management, approval workflows, and grant analytics. Layered Architecture separates the user interface, business logic, and data storage, making these responsibilities easier to organize, understand, and maintain.

Reason 2 - Suitable for a Centralized Academic Portal

Faculty Researchers, Co-authors, and the Research Dean access a common portal and shared data. Requests can move from the Presentation Layer to the Business Layer and then to the Data Layer in a clear and controlled manner, which suits the centralized nature of the system.

Security Advantage

Validation, authorization, and audit controls can be enforced in the Business Layer before changes reach the Data Layer. This helps protect sensitive grant and publication operations and supports secure handling of financial approvals and status modifications.

Performance Benefit

Separating business logic from data access allows database queries and frequently accessed grant, publication, and analytics information to be optimized independently. This supports responsive access under normal and peak usage without adding the operational complexity of a microservices architecture.